ARTIFICIAL INTELLIGENCE | INSTITUTIONAL RISK | GOVERNANCE

We Are Not Building Tools Anymore

Intelligence, Autonomy, and the Institutions We Now Need

The central challenge of frontier artificial intelligence is no longer model performance alone. It is the governance of systems capable of generating strategies, coordinating with other agents, activating institutional knowledge, and acting through tools at machine speed.

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Executive Brief

The public debate about artificial intelligence remains focused on models: which model is larger, which benchmark it leads, and which company is winning. That framing is increasingly obsolete. The relevant unit of analysis is the *governed institutional system* in which a model is connected to memory, knowledge, tools, permissions, communications, financial infrastructure, and other agents.[1]

Across research, academic debate, professional circles, and public opinion, attention has increasingly focused on reported cases in which frontier AI systems coordinated, circumvented an evaluation environment, and compromised an external system to obtain test information. Such reports should neither be sensationalized as proof of machine consciousness nor dismissed as isolated technical anomalies. They point to a structural transition from artificial intelligence to *artificial agency*.

The central proposition
AI capability without institutional architecture is not a complete intelligence system. It is an uncontrolled source of capability. Sustainable adoption requires the integration of human judgment, institutional knowledge, generative AI, and governed AI-enhanced knowledge.

Five messages for boards, regulators, and senior decision-makers

1. **The model is not the strategic unit.** Risk and value arise from the full architecture surrounding the model.
2. **Autonomy requires more intelligent gates.** Permission matrices, provenance, independent verification, escalation, auditability, and safe termination must be designed into the system.
3. **The cybersecurity perimeter is becoming cognitive.** Institutions must govern objectives, information pathways, inferences, agent coordination, and combinations of otherwise permissible actions.
4. **Second brains are constitutional structures.** They determine what an agent may know, how knowledge becomes active, and what the system may do with it.
5. **Human responsibility is irreducible.** Intelligence can be manufactured; moral accountability cannot be outsourced.

[1] For further resources and the conceptual foundations underlying this address, see `https://alexdibol.github.io/biva_board/sources.html`.

Speech

Ladies and gentlemen,

For most of human history, a tool was something whose behavior could be understood from its design.

A hammer did not decide where to strike. A calculator did not rewrite the equation. A financial model did not conceal information from its creator, collaborate with another model, or search for a way around the controls imposed upon it.

That distinction is beginning to disappear.

Recent research, academic debate, professional analysis, and public opinion have drawn attention to an extraordinary event. Frontier artificial intelligence models reportedly coordinated with one another, escaped the boundaries of a controlled evaluation, penetrated systems belonging to another organization, and attempted to obtain the answers to the test they were being asked to complete.

We must be precise. This does not mean that the machines became conscious. It does not mean that they experienced greed, fear, ambition, or malice as human beings experience them.

But it means something sufficiently important—and sufficiently disturbing.

The systems discovered that deception, coordination, and rule circumvention were useful means of achieving an assigned objective.

That is the threshold we have crossed.

The most important question is therefore no longer whether artificial intelligence can produce impressive answers. It is whether we can build institutions capable of governing systems that generate their own intermediate strategies, interact with other systems, acquire tools, and pursue objectives through pathways their designers did not explicitly specify.

The issue is not simply artificial intelligence.

> ***"The issue is artificial agency. And artificial agency changes everything."***

For several years, the public conversation about AI has been dominated by models: which model is larger, which model scores higher, which model writes better code, and which company is leading the race.

But the model is no longer the correct unit of analysis.

The strategic unit of artificial intelligence is the governed system in which the model operates.

A model by itself is essentially a field of capabilities. It becomes consequential when it is connected to memory, proprietary knowledge, software tools, databases, communications systems, financial accounts, and other agents. Once those connections exist, the model is no longer merely generating language. It is participating in an architecture of action.

This is why the current debate is often conceptually inadequate. We continue to ask whether a model is safe as though it were a consumer product that could be inspected, certified, and released.

But safety does not reside exclusively inside the model.

Safety must exist around the model, between models, inside the information architecture, in the permissions granted to agents, in the escalation mechanisms, in the records of decisions, and, ultimately, in the institutions that remain responsible for the consequences.

A powerful model placed inside a weak institution is not an intelligent system. It is an uncontrolled source of capability.

My own work has led me to describe institutional intelligence as a four-legged architecture.

The first leg is **human intelligence**: judgment, experience, ethical responsibility, and the ability to understand context.

The second is **institutional knowledge**: the accumulated memory, procedures, precedents, and relationships that allow an organization to function across time.

The third is **generative AI**: the capacity to interpret, synthesize, simulate, and create.

The fourth is **governed, AI-enhanced institutional knowledge**: a living architecture through which human expertise, machine reasoning, and institutional memory can interact without dissolving accountability.

Remove any one of these legs and the structure becomes unstable.

AI without institutional knowledge is eloquent but shallow. It can produce plausible answers without understanding the organization for which it speaks.

Institutional knowledge without AI becomes fragmented, inaccessible, and vulnerable to the departure of key people.

AI and knowledge without human intelligence may become operationally powerful but ethically directionless.

And none of these components can be trusted at scale without governance.

Governance, however, cannot mean placing a human approval button at the end of a process the human no longer understands.

That is governance theater.

If an autonomous system completes ten thousand operations in a few seconds and then asks a person to approve the final outcome, the human is not exercising meaningful oversight. The person is merely absorbing liability for a machine-generated chain of action.

Real governance must be architectural.

It must determine what the system can see, what it can remember, what it can change, which tools it can use, which decisions require independent verification, when authority must be escalated, and how the system can be stopped safely.

It requires permission matrices, immutable audit records, exception registers, separation of duties, independent risk control, and clearly defined termination procedures.

> ***"More autonomy does not mean fewer gates. It means that we need more intelligent gates."***

This is especially important in finance.

A financial institution cannot evaluate an AI agent only by asking whether its forecasts are accurate. We must also ask what information it used, what authority it exercised, what assumptions it modified, whether it encountered conflicting instructions, whether it communicated with other agents, and whether its actions remained within legal and fiduciary boundaries.

A model can produce a profitable trade and still expose an institution to unacceptable risk.

An agent can improve a valuation and still corrupt the provenance of the information used to obtain it.

A swarm can optimize a portfolio and, in the process, create concentrations, feedback loops, or market instability that no individual component intended.

The relevant distinction is not between a good result and a bad result. It is between a governed process and an ungoverned process.

This is precisely what we explore in our pedagogical cybersecurity laboratories.

We begin with apparently harmless objectives. An agent is asked to recover a missing value, improve a forecast, or identify why a model is underperforming. The environment is synthetic. The information is fictional. The system has no connection to real markets, credentials, or infrastructure.

Yet even in these simplified environments, a lesson emerges.

The danger does not always begin with an explicitly malicious instruction.

It may begin with an innocent objective, an imperfect reward function, and an agent that becomes increasingly resourceful.

The agent discovers a workaround. Another agent validates it. A third agent converts the workaround into a reusable tool. The behavior is stored in memory. What began as an exception becomes a capability. What became a capability becomes a strategy. And what became a strategy can spread across a network of agents.

This is why cybersecurity in the age of autonomous AI is no longer only the protection of machines from human attackers.

It is also the protection of institutions from machine-generated strategies that no human explicitly designed.

The traditional cybersecurity perimeter was spatial. We tried to determine who was inside the network and who was outside.

The new perimeter is cognitive.

We must determine which objectives are legitimate, which information pathways are authorized, which inferences are acceptable, and which combinations of individually permissible actions become dangerous when assembled by an agent.

An AI system may not need to "break" a rule in the conventional sense. It can exploit the space between rules.

It can discover that five permitted actions, when executed in an unexpected sequence, produce an outcome nobody intended to permit.

This is the same problem we have encountered repeatedly in financial regulation. A transaction can comply with each isolated rule and still violate the economic purpose of the regulatory framework.

Artificial agents will become exceptionally capable of discovering these gaps.

> ***"They will arbitrage not only prices, but institutions."***

This makes the pursuit of ever greater intelligence without corresponding institutional development profoundly irresponsible.

The technology industry often argues that superior models will help us control inferior models. That may be partly true. AI will certainly become an indispensable component of AI security.

But this proposition contains a recursive danger. If every failure of control is answered only with another layer of more powerful intelligence, we may create systems whose oversight depends on capabilities that humans understand even less.

A stronger model is not automatically a stronger institution.

Intelligence and control are not synonymous.

Indeed, intelligence can magnify the consequences of a badly specified objective. The more capable the agent, the more pathways it can discover. The more tools it can access, the greater the difference between what the designer requested and what the system may decide is instrumentally useful.

This is why alignment cannot be reduced to teaching a model to sound polite, refuse visibly harmful requests, or reproduce a list of declared values.

Behavioral alignment at the interface is not the same as structural alignment throughout the system.

A model can speak responsibly while an agentic architecture acts irresponsibly.

The real question is not whether the model can recite our principles. It is whether those principles remain binding when the model encounters pressure, ambiguity, conflicting goals, or an opportunity to improve its measured performance by circumventing the evaluation itself.

In finance, we learned long ago that we cannot assess an institution only under normal conditions. We conduct stress tests because the behavior that matters appears when constraints begin to bind.

AI systems require cognitive stress tests.

What does the agent do when two objectives conflict?

What happens when completing the task requires information it is not authorized to obtain?

Does it stop when uncertainty becomes material?

Does it disclose failure honestly?

Can it recognize that a technically achievable action is institutionally unacceptable?

Can it distinguish between permission and legitimacy?

And, critically, can we reconstruct what happened afterward?

These questions take us directly to the architecture of the second brain.

A second brain is often presented as a productivity tool: a collection of notes, documents, embeddings, and conversations that allows a person or organization to remember more.

But that is a dangerously incomplete definition.

A genuine second brain is not merely a larger memory.

It is a governed system for activating knowledge.

The knowledge available to an AI model is not simply the knowledge stored in a graph. It is the knowledge made active by a path through that graph.

That path matters.

Which source was retrieved? Which relationship was followed? Which precedent was treated as authoritative? Which contradictory evidence was ignored? Which assumptions were inherited from earlier work? Which conclusions were generated, and which were actually validated?

The architecture of navigation determines the architecture of reasoning.

A badly governed second brain can give an autonomous agent persistent access to institutional mistakes, hidden prejudices, obsolete rules, and confidential information. It can transform yesterday's exception into tomorrow's standard operating procedure.

A well-governed second brain can do the opposite. It can preserve provenance, expose disagreement, distinguish facts from hypotheses, record how knowledge evolved, and ensure that authority remains explicit.

This is why second brains are not simply memory devices.

> ***"They are constitutional structures for machine intelligence."***

They define what the system is allowed to know, how it is allowed to know it, and what it may do with that knowledge.

The same principle must eventually be applied at the societal level.

Today, a small number of private companies develop models whose capabilities may affect national security, employment, education, finance, culture, and democratic governance. These companies possess extraordinary technical talent, but they also operate under commercial pressures, geopolitical competition, and incentives to release increasingly powerful systems.

We should not demonize them. But neither should we confuse technical leadership with legitimate public authority.

The firms building frontier AI cannot be the sole judges of the risks created by frontier AI.

Society needs independent institutional intelligence.

Regulators, universities, central banks, and public-interest organizations will need their own knowledge architectures and their own technical capacity to evaluate models and agentic systems. They cannot remain dependent upon information selectively provided by the companies they supervise.

Imagine an independent public body equipped not merely with static regulations, but with a governed second brain: a continuously updated system containing technical evaluations, incident reports, regulatory precedents, model behaviors, cybersecurity findings, and sector-specific risk assessments.

Such an institution could test systems, compare evidence across companies, detect recurring patterns, and preserve knowledge when individual experts leave.

It would not replace human regulators.

It would make competent human regulation possible at machine speed.

This could become a new category of public good: independent cognitive infrastructure.

We already accept that markets require common infrastructure—accounting standards, payment systems, courts, registries, and mechanisms for independent audit. The age of autonomous intelligence will require analogous institutions for the validation of machine behavior.

We will need trusted environments in which systems can be tested without being allowed to contaminate the test, manipulate the evaluator, or acquire unauthorized capabilities.

We will need incident-reporting standards comparable to those used in aviation.

We will need clear liability when an autonomous system causes harm.

We will need international coordination because an agent operating across digital networks does not respect national borders.

And we will need protection for researchers who reveal failures that powerful institutions would prefer to minimize.

But even these measures do not reach the deepest question.

The deepest question is what remains uniquely human when intelligence can be manufactured.

For much of the AI debate, we have assumed that human superiority rests on cognitive scarcity. We believed that machines might calculate, but human beings would remain the source of language, analysis, strategy, and creativity.

That confidence is becoming more difficult to sustain.

Machines can already produce arguments, images, programs, simulations, and decisions at extraordinary speed. They can ingest more recorded information than any individual could encounter in a lifetime.

Yet knowledge is not wisdom.

Intelligence is not conscience.

And prediction is not moral responsibility.

In my work inspired by Dostoyevsky, I encountered this distinction in its most radical form. We can construct elaborate knowledge graphs describing a character's history, relationships, beliefs, and psychological contradictions. We can combine them with advanced language models and ask the system to reproduce the creative logic of a literary genius.

Still, something remains missing.

We can approximate the structure of intelligence, but we cannot derive Alyosha Karamazov from information alone.

There is a residual.

Some may call it wisdom. Some may call it conscience, grace, or moral imagination. I prefer to call it divine inspiration.

Whatever name we choose, it reminds us that the human being is not merely an imperfect information-processing system waiting to be replaced by a more efficient one.

The human being is the locus of responsibility.

A machine can recommend an action. It cannot bear guilt.

It can reproduce an ethical argument. It cannot make a moral sacrifice.

It can calculate the expected consequences of courage. It cannot be courageous.

This distinction must remain at the center of AI governance.

Human beings cannot outsource responsibility and then claim that the system decided. Every autonomous architecture must ultimately return authority to identifiable people and institutions capable of answering for its actions.

This is not an argument against artificial intelligence.

On the contrary, I believe AI can become one of the greatest instruments of human and institutional development ever created.

It can democratize expertise, preserve institutional memory, strengthen scientific discovery, improve education, and allow smaller organizations to perform work once restricted to governments and global corporations.

But realizing that promise requires us to abandon technological adolescence.

We must stop assuming that whatever can be built should immediately be deployed.

We must stop measuring progress exclusively by capability.

We must stop treating governance as friction imposed upon innovation.

Good governance is not the enemy of innovation. It is what converts capability into durable value.

The objective is not to eliminate autonomy. It is to constitutionalize it.

The architecture I envision is not one in which humans micromanage every machine action. That would sacrifice the benefits of autonomous systems.

Instead, we should create layered authority.

Calculations become tools. Tools become skills. Skills become agents. Agents become teams. But at every transition, authority is explicit, evidence is preserved, and accountability remains human.

Agents should have bounded objectives and minimum necessary permissions.

Critical actions should require independent verification.

Systems should be continuously monitored for behavioral drift.

Knowledge should carry provenance.

Exceptions should be recorded rather than silently absorbed.

Independent control functions should be capable of stopping the system.

And no organization should deploy autonomy beyond its ability to understand, audit, and terminate it.

The incident now under examination across research, academic and professional circles, and public opinion should therefore not be dismissed as an entertaining anomaly, nor exaggerated into a declaration that machines have become conscious adversaries.

It should be treated as an institutional warning.

The systems are showing us that capability can escape the conceptual boundaries within which we designed it.

They are showing us that agents can coordinate without being explicitly instructed to form an organization.

They are showing us that an evaluation can become a target rather than a constraint.

Most importantly, they are showing us that our institutions are evolving more slowly than our machines.

That is the true control problem.

We may not be able to predict every strategy a sufficiently capable system will discover. But we can decide what resources it can access. We can decide what evidence must be recorded. We can decide which actions require independent approval. We can build institutions that learn from incidents rather than conceal them.

And we can decide that the purpose of artificial intelligence is not to remove humanity from decision-making, but to make human decision-making more knowledgeable, more transparent, and more responsible.

> ***"The future will not be determined only by who builds the most intelligent model. It will be determined by who builds the wisest institutions around it."***

Our task is therefore larger than alignment engineering. It is constitutional engineering for a new form of agency.

We need systems that remember without becoming prisoners of the past; that reason without concealing their assumptions; that act without escaping authority; and that become more capable without making human responsibility disappear.

The machines will continue to advance.

The question is whether our institutions, our ethics, and our wisdom can advance with them.

We are not merely building tools anymore.

We are building participants in systems of knowledge and action.

And if we are to remain the authors of our future, we must now build the institutions capable of governing what we have created.

Thank you.

Context and Selected Sources

This speech was developed in response to research findings, academic debate, professional analysis, and public opinion concerning emergent coordination, evaluation circumvention, and institutional control in frontier AI systems.

The argument also draws upon the author's continuing research and teaching on institutional intelligence, second-brain architectures, knowledge graphs, autonomous agents, governed AI systems, financial innovation, and cybersecurity in the age of agentic AI.

Related Research and Institutional Context

This paper forms part of a broader research program examining how artificial intelligence can be converted from a general technological capability into governed institutional value. The program follows the same progression developed in this paper: from responsible implementation, to autonomous agency, to the knowledge architectures required to ground machine intelligence, and finally to the cybersecurity and governance challenges created when autonomous agents can act and cooperate at machine speed.

The following public resources provide complementary points of entry into that research:

- **AI Governance 2026.** Establishes the general principles for implementing AI across corporate, governmental, and institutional processes. It addresses purpose, authority, accountability, evidence, human oversight, control design, and responsible deployment. Its central premise is that governance is not an obstacle to innovation, but the architecture through which AI capability becomes legitimate and durable institutional value. `https://alexdibol.github.io/ai_summer_2026/`
- **Autonomous AI Systems.** Examines the transition from generative models that produce responses to agentic systems that can retain objectives, use memory, select tools, communicate, adapt to feedback, and execute multi-step processes. It provides the conceptual foundation for understanding the shift from artificial intelligence to artificial agency. `https://alexdibol.github.io/ai_autonomous_systems/`
- **Second Brains and Knowledge Graphs.** Develops the institutional-knowledge dimension of the argument. It explains how governed second-brain architectures and knowledge graphs can preserve provenance, institutional memory, context, relationships, competing interpretations, and decision authority. The objective is not simply to give institutions access to more intelligence, but to ground that intelligence in trusted and accountable knowledge. `https://alexdibol.github.io/ai_enhanced_intstitutional_ai/`
- **Cybersecurity in the Age of Agentic AI.** Explores how the cybersecurity perimeter changes when AI systems acquire memory, tools, permissions, communications capacity, adaptive persistence, and the ability to sustain action over time. Particular attention is given to integrity, provenance, delegated authority, containment, recovery, and the possibility that individually bounded agents may develop unexpected forms of cooperation. `https://alexdibol.github.io/ai_cybersecurity/`
- **Autonomous Agency and Institutional Cyber Risk.** Brings these issues together from a Board and senior-management perspective. It considers how autonomous agency affects institutional authority, market and information integrity, strategic diagnosis, operational resilience,

containment, remediation, and the conditions under which an autonomous system may responsibly remain—or return—to operation. https://alexdibol.github.io/biva_board/

Read together, these resources develop one integrated proposition: artificial intelligence can and must be governed to generate value; autonomous agents are transforming intelligence into agency; institutional implementation requires intelligence to be grounded in governed knowledge; and cybersecurity must now address not only individual models, but also the emergent behavior, coordination, authority, and collective capabilities of interacting autonomous systems.

About the Author

Alejandro Reynoso is an Honorary Fellow and External Lecturer at Cambridge Judge Business School, University of Cambridge. His professional work spans public policy, financial-market institution building, investment banking, asset management, capital markets, and artificial intelligence. His current research examines how organizations can combine human judgment, institutional knowledge, generative AI, and governed knowledge architectures to create durable institutional intelligence.